

Figure 1 — Defining the drug-indication approval-forecast task

A Prediction-unit comparison

Prior task	Prediction unit	Limitation
LOA analysis	disease / phase / modality rate	not individualized
Drug-level approval	drug	not disease-specific
Trial-outcome prediction	trial	not final approval, often trial-feature dependent

This work drug-indication pair → individualized, disease-specific, approval-level (trial-agnostic)

B Dataset construction pipeline



C Final cohort summary

14,134 drug-indication pairs	2,469 / 11,665 approved / failed	2,637 test pairs (624 positive)
2,313 unique drugs	4,190 unique indications	5.18 avg indications / drug